25901205 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSECFS201 - LINUX ADMINISTRATION

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Associate system users with their purpose in Linux. [U] / [CO1]
2. Identify the main types of files in Linux. [U] / [CO1]
3. Indicate the role of the file system in Linux. [U] / [CO1]
4. Write a command to calculate 25 × 4 using expr. [A] / [CO2]
5. Apply the find command to locate a file named report.txt. [A] / [CO2]
6. Define symbolic mode in chmod. [K] / [CO2]
7. Compare the roles of cd and pwd in directory management. [An] / [CO2]
8. Define loop control structures. [K] / [CO3, CO4]
9. Write a shell program to display a given string. [A] / [CO3, CO4]
10. List the keywords in shell. [K] / [CO3, CO4]
11. Describe the role of RPM in Red Hat Linux. [U] / [CO3, CO4]

12. Justify any one practical application of DNS in daily internet use. [E] / [CO3, CO4]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

13. Categorize process management commands into monitoring and controlling tools, and justify your classification with examples. [An] / [CO2]

14. Explain the working of the head, tail, and wc commands with suitable examples. [U] / [CO2]

15. Explain the purpose and usage of the ifconfig command in Linux. How can it be used to check the IP address of a system. [U] / [CO2]

16. Explain the concept of mounting and unmounting file systems in Linux. Describe the role of the mount and umount commands with suitable examples. [U] / [CO2]

17. Explain any four positional parameters with their meanings. [U] / [CO3, CO4]

18. Describe any 5 text entry commands in vi editor. [U] / [CO3, CO4]

19. Justify the preference using sudo instead of working directly as the root user in multi-user Linux systems. [E] / [CO3, CO4]

20. Explain how Linux differentiates access rights between file owners, groups, and others. [U] / [CO3, CO4]

[5x6 = 30]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

21. Compare Bourne Shell, Bash Shell and C Shell. [U] / [CO1]

22. Explain different types of shell available in linux with their features. [U] / [CO3, CO4]

23. Evaluate the effectiveness of managing users and groups separately in Linux. Explain how this separation improves system security and simplifies administration in multi-user environments. [E] / [CO3, CO4]

[10x2 = 20]